

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT3 — CASE STUDY

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO2 Statement:	<i>Analyze real-world data scenarios and evaluate/justify the most appropriate chart types, coordinate systems, color scales, and dashboard designs to represent them.(BL5)</i>		
Student Name:	Nivya Sri T	Reg. No.:	192424131
Section / Batch:	2024	Date:	01-08-26

Code of Conduct

I Nivya Sri T (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nivya Sri. T

Instructions

- This test contains 2 case-study questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each case study has four sub-parts (a–d); answer all parts for full marks.
- Support your answers with brief calculations or justifications where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – CASE STUDY

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Case Study

CO2 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO2-AT3 Case Study questions, in which students analyze a realistic multi-part scenario and select, justify, design, and critique appropriate visualizations across chart types, coordinate systems, color scales, and dashboard development.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the full range of chart types, coordinate systems, and color scales relevant to the case, and correctly identifies which apply to each sub-part.	Good understanding of the relevant concepts, with only minor gaps in identifying appropriate chart types, scales, or color choices.	Basic understanding, but with some misconceptions about which visualization techniques suit the scenario's data.	Poor understanding of core concepts; proposes techniques fundamentally unsuited to the case study's data.
Explanation	25%	Provides detailed, well-reasoned justification for every design decision and critique, explicitly tied to specific details of the case.	Provides clear justification for most decisions and critiques, with adequate reasoning.	Justification is limited or generic; decisions are stated without sufficient reasoning tied to the case.	Little or no justification provided; answers are unsupported assertions.
Application	20%	Effectively applies visualization concepts to analyze and solve every sub-part of the case, including any calculations, with well-reasoned conclusions.	Applies concepts correctly to most sub-parts, with only minor errors.	Limited application; some sub-parts are weakly addressed or only partially correct.	Fails to apply appropriate concepts; sub-parts are left unaddressed or answered incorrectly.
Organization	15%	The response is clearly structured, addressing	The response is organized and addresses each sub-part, with	Basic structure; sub-parts are addressed but the response	Poorly organized; the response is incoherent, and

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		every sub-part (a–d) in a logical sequence with coherent overall flow.	only minor issues in flow or clarity.	lacks clarity or a logical sequence.	sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — chart choices, calculations, critiques, and dashboard design — are completely accurate and well-suited to the case.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the case, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO2 AT3: CASE STUDY

1.

a) What axis scale is needed for the holdings data, and why is a linear scale unsuitable here?

A logarithmic axis scale is required for holdings when the values span large magnitudes (for example, thousands vs millions).

A linear scale is unsuitable because:

- Smaller holdings get compressed near zero and become visually insignificant.
- Larger holdings dominate the chart, hiding meaningful variation.
- Proportional differences are not visible.
- Log scale spreads values evenly by magnitude, revealing true growth patterns.

b) Design a chart for monthly returns that clearly distinguishes gains from losses using color.

Use a diverging bar chart.

Design

- X-axis: Month
- Y-axis: Monthly Return (%)
- Bars extend upward for gains and downward for losses.
- Green for positive returns.
- Red for negative returns.

Why this works

- Gains vs losses are instantly visible.
- Color reinforces direction.
- Magnitude of volatility is easy to compare.
- This is the standard financial visualization for return series.

c) Propose a chart for the 6-category asset allocation and justify your choice over at least one alternative.

Use a horizontal stacked bar chart.

Why this is the best choice

- Shows proportions clearly across all 6 categories.
- Easy to compare category sizes.
- Works well when categories exceed four.
- Labels and percentages fit cleanly without clutter.

Alternative (and why it's weaker): Pie Chart

- Difficult to compare six slices accurately.
- Human perception struggles with angles.
- Labels overlap and reduce readability.
- Not ideal for multi-category financial allocation.

d) Combine two of the above into a coordinated dashboard, with an annotation highlighting the riskiest month.

Dashboard Components

1. Monthly Returns Diverging Bar Chart
 - Shows gains and losses with green/red bars.
 - Highlights volatility.
2. Asset Allocation Stacked Bar Chart
 - Shows portfolio composition.
 - Helps interpret exposure during risky periods.

Coordination

- Use Month as a filter or highlight action.
- Selecting a month in the returns chart can highlight related allocation exposure.

Annotation

Identify the riskiest month — the one with the largest negative return.

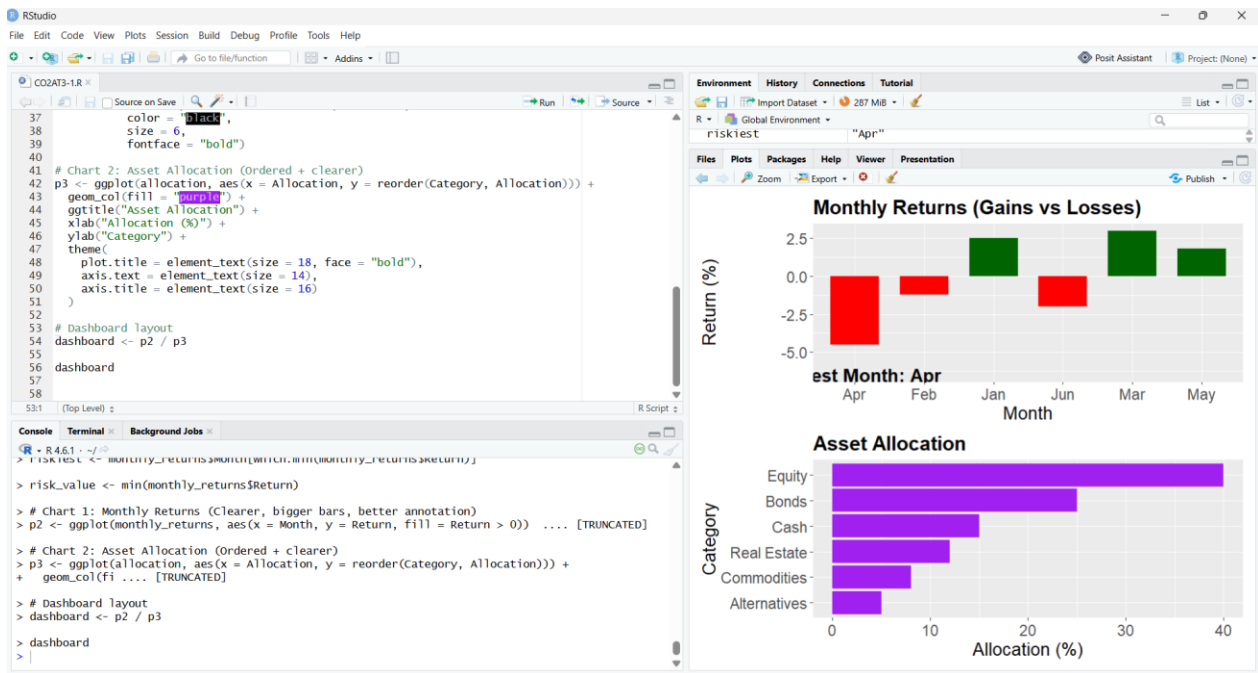
Add an annotation such as:

“This month shows the highest downside volatility — review exposure to high-risk asset classes.”

Dashboard takeaway

- The returns chart reveals when losses occurred.
- The allocation chart shows which asset classes dominated during that period.
- Together, they help identify whether concentration risk contributed to the downturn.

VIZUALIZATION:



2.

a) Propose a chart combining headcount and attrition rate for the 6 departments, with an annotation on the department needing urgent retention action.

- Bars → Department headcount
- Line → Attrition rate (%)
- X-axis → Departments
- Left Y-axis → Headcount
- Right Y-axis → Attrition rate

This chart works because headcount and attrition are on different scales, and the dual-axis layout allows both to be compared without distortion.

Annotation

Add a text callout on the department with the highest attrition rate:

“Urgent retention action needed: Highest attrition observed in [Department X].”

This makes the risk immediately visible.

b) Which chart best reveals the shape, median, and outliers of the company-wide salary distribution?

The best chart is a boxplot.

A boxplot clearly shows:

- Shape → via spread and whiskers
- Median → the central line inside the box
- Outliers → plotted as individual points
- Distribution symmetry → via box and whisker lengths

No other chart (histogram, violin, density) shows median + quartiles + outliers as cleanly in one view.

c) Which coordinate system and chart type is most natural for the fixed-total, 40-hour workweek time allocation? Justify your answer with an angle calculation for an 8-hour component.

Use a polar coordinate system with a pie chart (or donut chart).

Why?

- A 40-hour workweek is a fixed total, making angle-based representation natural.
- Pie charts encode parts of a whole using angles.

Angle calculation for an 8-hour component

Total = 40 hours Component = 8 hours

$$\text{Angle} = \frac{8}{40} \times 360^\circ = 72^\circ$$

So an 8-hour task occupies 72 degrees of the pie.

This demonstrates why the pie chart is intuitive: the viewer immediately sees proportional time allocation.

d) A manager wants to use 3D pie charts for the dashboard “because they look more impressive.” Explain the visualization problems this introduces.

3D pie charts introduce several visualization issues:

1. Distorted angles The 3D tilt makes slices in the front appear larger than slices in the back, even if values are equal.
2. Incorrect perception of area The 3D perspective stretches some slices, misleading the viewer about actual proportions.
3. Hidden or partially obscured slices Smaller categories can become visually hidden behind larger ones.
4. Harder comparison Human perception already struggles with angle comparison; 3D makes it worse.
5. “Impressive” but inaccurate They look flashy but reduce clarity and analytical correctness.

In dashboards, accuracy > aesthetics, so 3D pie charts should be avoided.

VISUALIZATION:

